

ZAIN TAMER ZAIN ELABDIN

Machine Learning & AI Engineer

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Summary

Computer Engineering student with hands-on experience building and deploying end-to-end machine learning solutions for computer vision, NLP, and predictive analytics. Experienced in developing production-ready ML pipelines, cloud deployment, and MLOps using Azure, AWS, Docker, and MLflow. Passionate about applying modern AI techniques to solve real-world business problems.

Education

Arab Academy for Science, Technology & Maritime Transport University

Bachelor of Science in Computer Engineering

Alexandria, Egypt

Sep 2022 – Feb 2027

CGPA: 3.98/4.0

Relevant Coursework: Probability & Statistical Analysis, Artificial Intelligence, Data Analytics and Optimization, Database Systems, Object-Oriented Programming, Data Structure & Algorithms, Distributed and Parallel Systems.

Experience

Machine Learning Engineer Intern – Digital Egypt Pioneers Initiative

Jun 2025 – Dec 2025

- Completed a 180-hour applied ML program covering computer vision, NLP, data engineering, and cloud deployment.
- Designed, trained, and deployed end-to-end machine learning pipelines for computer vision and predictive analytics, covering **data preprocessing, feature engineering, model training, evaluation, and cloud deployment** on **Azure Machine Learning** and **Azure App Service**.
- Developed NLP pipelines for text classification and sentiment analysis, applying Transformer-based models to real-world text datasets and managing artifacts on **Azure Blob Storage**.

Projects

RAG Knowledge Base Chatbot – Production-Ready LLM Application

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** chatbot using **LangChain**, **Hybrid Search (BM25 + FAISS)**, and **HuggingFace Embeddings** to answer questions over a custom knowledge base.
- Developed and deployed a production-ready backend with **FastAPI**, **Docker**, and **AWS EC2**, exposing REST APIs for question answering and integrating **LangSmith** for LLM tracing.

Premier League Predictor – End-to-End MLOps System

- Built an end-to-end machine learning system to predict English Premier League match outcomes using historical team statistics and engineered features.
- Designed a reproducible **MLOps** workflow using **ZenML**, **MLflow**, **Docker**, **AWS S3**, **PostgreSQL**, **EC2**, and **ECR** for automated training, experiment tracking, deployment, and weekly model evaluation.

FPL Vision – AI-Powered Fantasy Premier League Assistant

- Built a full-stack web assistant for Fantasy Premier League managers, generating player recommendations and expected points projections using a trained **XGBoost** model.
- Processed data from the FPL API (700+ players, 20 teams) and deployed the application on **Azure App Service** with **Azure Blob Storage** and **GitHub Actions CI/CD**.

Facial Recognition System

- Developed a real-time facial recognition application using FaceNet embeddings, transfer learning, and MediaPipe multi-face detection, achieving 98% recognition accuracy.
- Deployed the Dockerized **Streamlit** application on Hugging Face Spaces.

Skills

Programming Languages: Python, C, C++, C#, Java, SQL, Bash

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, Transformers, LangChain, Retrieval-Augmented Generation (RAG), FAISS, BM25, HuggingFace Embeddings, OpenCV, Pandas, NumPy, Matplotlib, Plotly, Deep Learning

Backend & Databases: FastAPI, PostgreSQL, REST APIs

Cloud & MLOps: Amazon Web Services (EC2, ECR, S3), Docker, ZenML, MLflow, LangSmith, Git, GitHub, GitHub Actions, Linux, Hugging Face Spaces

Certificates

Second Place In HackerRank x CPCLub AAST Event	Apr 2026
Codeforces Specialist (max rating: 1448)	Nov 2025
ARTIFICIAL INTELLIGENCE ENGINEER 1 (Coursera – IBM)	Oct 2025
Sprints x Microsoft Summer Camp – AI and Machine Learning	Oct 2025
ECPC (Egyptian Collegiate Programming Contest) Qualifications	Jul 2025
15th Place in Zindi Financial Inclusion in Africa Competition	Jun 2025
IBM Software Developer Roadmap (Coursera – IBM)	Oct 2024
IEEEExtreme Programming Competition	Oct 2024
• Ranked in top 2% in the Egyptian region and in top 20% worldwide from over 8000 teams.	

Volunteering

AWS Student Builder Group at AAST - Team member	Mar 2026 - Present
• Participated in organizing several cloud practitioner training sessions. • Lead a workshop on different AWS services for 20+ students.	
AAST Competitive Programming Club – Coach	Sep 2025 – Present
• Led 15+ tutoring sessions explaining competitive programming fundamentals and algorithms. • Monitored and coached a group of 30+ students. • Contributed in preparing materials and contests for trainees.	